

Professional Summary	
Data Scientist & AI Engineer with expertise in Machine Learning, Deep Learning, NLP, Computer Vision, Data Engineering, and Business Intelligence. Experienced in end-to-end data projects including data collection, preprocessing, modeling, deployment, and dashboard development. Passionate about solving complex problems with AI and data-driven solutions.	
Technical Skills	
Programming & Tools	Python (advanced), R, SQL, MongoDB, Cassandra, Git/GitHub, FastAPI, React.js, Django, Bootstrap
Data Engineering & Big Data	ETL pipelines, PySpark, Spark MLlib, Hadoop, Data Warehousing, Dimensional Modeling, NoSQL, AWS, Azure, GCP (basic)
Machine Learning & AI	Supervised & Unsupervised Learning, Predictive Modeling, Deep Learning (ANN, CNN, RNN, LSTM, GRU), NLP (Transformers, BERT, GPT, Seq2Seq, Attention, Embeddings, RAG, Text Mining), Computer Vision (YOLO, UNet, Mask R-CNN, Transfer Learning), Generative Models (GAN, VAE, Diffusion Models), MLOps, Model Deployment (FastAPI)
Business Intelligence & Visualization	Power BI, Tableau, Dashboards, Reporting, Business Analysis
Statistics & Analysis	Advanced Statistical Analysis, Regression, Time Series, Feature Selection, Predictive Modeling, Model Evaluation
Projects	
Sentiment Analysis (2025)	NLP model for sentiment analysis using Python. Tools: Python, NLTK, Scikit-learn.
Price Prediction from Images & Descriptions (2024-2025)	Multimodal AI model combining image features and text embeddings. Tools: Python, PyTorch, HuggingFace, OpenCV.
Throat Cancer Detection (2024-2025)	Classification model for medical diagnosis using R. Techniques: Feature selection, Machine Learning models.
Carpooling Management App (2024-2025)	Full-stack web application with backend APIs and frontend interface. Tools: Django, Bootstrap, MySQL.

Data Engineering & BI Dashboards	Designed ETL pipelines, created interactive dashboards and reporting solutions. Tools: Power BI, Tableau.
Smart Home Automation with Spark ML	Predictive analytics and anomaly detection for smart home devices. Tools: Spark ML, Python.
Fraud Card Detection Pipeline (MLOps)	End-to-end pipeline for detecting fraudulent credit card transactions. Tools: Python, Scikit-learn, MLflow, Docker, CI/CD.
Tree Leaf Disease Detection (2025)	Real-time disease detection using transfer learning and YOLO with camera input. Tools: Python, PyTorch, OpenCV, YOLO.

Work Experience

- 2025 **Developer, *BigDeal***
Developed a conversational agent for customer support in Privilege Club.
- 2024 **Developer, *Tunisair***
Web development of an internship management application.
- 2023 **Developer, *InnovUp (PFE)***
Developed investment platform, including to-do planning, project diagrams, and overall implementation.

Education

- 2024–Present **Engineering Degree in Data Science & AI, *Tek-up University*, Tunisia**
- 2020–2023 **Bachelor's Degree in Computer Science, *Institut Supérieur d'Informatique*, Tunisia**
- 2018–2019 **Baccalauréat in Technical Sciences, *Chebbi High School*, Tunisia**

Languages

- French Fluent
- Arabic Native
- English Intermediate